Introduction to Pharmaceutical Science

Science

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Ollscoil Teicneolaíochta an Oirdheiscirt
South East Technological University

Introduction to Pharmaceutical Science (A09660)

Short Title:	Introduction to Pharma Science
Faculty	Science and Computing
Department	Science
Cluster	Pharmaceutical Science
Author	ADONEGAN
Credits	10
Level:	Introductory

Description of Module / Aims

This module covers several aspects of pharmaceutical science, including classification of pharmaceutical products, types of formulations, the manufacturing process and the role of the operator in this core value-added activity within a pharmaceutical organisation.

Programmes

`PHAR-0002`	Higher Certificate in Science in Good Manufacturing Practice and Technology (`WD_SGMPT_C`)	2 / 3 / M

Indicative Content

- Terminology, general requirements of pharmaceuticals and pharmaceutical manufacturing processes
- Classification of pharmaceuticals, topical, oral dose, parenterals, aerosol, OTC, prescription, sustained release products, biopharmaceuticals etc.
- Stages in pharmaceutical development
- Requirements of a quality product
- The manufacturing processes involved in the production of pharmaceuticals: blending, granulation, drying, micronising, encapsulation, tableting, coating
- Introduction to pre-formulation studies
- ICH guidelines
- Introduction to the role of excipients
- Requirements for stability testing on a pharmaceutical product

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Describe the requirements of pharmaceutical development and differentiate between different classes of pharmaceuticals.
2. Outline the pathway to be followed in the development of a pharmaceutical product.
3. Review ICH guidelines and apply their knowledge in drawing up a stability protocol for a pharmaceutical finished dosage form.
4. Explain the primary processes used in the manufacture of finished dosage forms.
5. Discuss a protocol that could be followed for a preformulation study on an API in advance of pharmaceutical formulation.

Learning and Teaching Methods

- Lectures.
- Academic journals.
- E-learning via Moodle.

Assessment Methods

	Weighing	Outcomes Assessed
Final Written Examination	50%	1,2,4,5
Continuous Assessment	50%	
Assignment	25%	3
In-Class Assessment	25%	1,2

Assessment Criteria

<40%: Very limited knowledge and understanding of the subject matter. Lacking in technical ability/skill and understanding to carry out practical/professional tasks. Failure to meet the objectives of projects. Unable to carry out or submit independent study and research.

40%-49%: Demonstrate a limited knowledge of the subject matter. Show some technical ability/skill in carrying out and reporting on practical/professional tasks. Attempt made to meet objectives and carry out independent study and research. Final result lacks balance and arguments undeveloped.

50%-59%: Demonstrate satisfactory general knowledge of the main issues within the subject. Logically and competently carry out practical/professional tasks.

60%-69%: Demonstrate sound knowledge of the subject matter. Show ability to analyse and logically argue in an effective and mature style. Demonstrate initiative and the ability to criticise methods used and appreciation of experimental design when carrying out practical/professional tasks. Projects and reporting coherent and soundly structured illustrating wide and in depth reading on subject coupled with the proper use of references.

70%-100%: Demonstrate authoritative handling of complex material and a highly developed and extensive understanding of the subject. Provide well focused analysis and convincing argument on the subject.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture		24
Independent Learning		246

Supplementary Material

- Liebermann, H.A. _Pharmaceutical Dosage Forms Tablets_ . New York: Marcel Dekker , 1981.